

Neutron reactions in the MX17 detector — products & γ energies (thermal capture $Q = S_n$ of product; AME2020/ENDF)

Material / volume	Reaction	Target (abund.)	Q [MeV]	Products (energy)	γ emitted	Role
He-3 gas	$^3\text{He}(n,p)^3\text{H}$	^3He	0.764	p 573 keV + t 191 keV	none (charged)	Beam monitor (dominant gas rxn)
He-3 gas	$^3\text{He}(n,\gamma)^4\text{He}$	^3He	20.578	20.58 MeV γ or e^+e^- pair	20.58 MeV (or IPC)	SIGNAL — X17 / IPC channel
Al capsule	$^{27}\text{Al}(n,\gamma)^{28}\text{Al}$	^{27}Al (100%)	7.725	γ cascade $\rightarrow$ Compton e^-	$\Sigma = 7.72$ MeV (1779 keV +..)	MAIN coincidence bkg (96% of legs)
LS / plastics / CFRP	$^1\text{H}(n,\gamma)^2\text{H}$	^1H	2.224	single 2.223 MeV $\gamma \rightarrow e^-$	2.223 MeV	Soft γ : many singles, few legs
plastic / LS (fast n)	$^1\text{H}(n,n')p$	^1H	elastic	recoil proton (direct)	none	Non- γ ionization bkg
CFRP / plastic	$^{12}\text{C}(n,\gamma)^{13}\text{C}$	^{12}C (98.9%)	4.946	γ cascade $\rightarrow e^-$	4.945, 3.089, 1.262 MeV	Minor γ bkg
PMT glass	$^{28}\text{Si}(n,\gamma)^{29}\text{Si}$	^{28}Si (92%)	8.473	γ cascade $\rightarrow e^-$	$\Sigma = 8.47$ MeV	Minor bkg
readout Cu	$^{63}\text{Cu}(n,\gamma)^{64}\text{Cu}$	^{63}Cu (69%)	7.916	γ cascade $\rightarrow e^-$	$\Sigma = 7.92$ MeV	Minor bkg
air (trace)	$^{14}\text{N}(n,p)^{14}\text{C}$	^{14}N (99.6%)	0.626	proton 584 keV	none	Trace

Detector hits are deposited by the secondary electrons of the capture γ (Compton/photoelectric), except the direct recoil proton from fast-neutron elastic scattering and the charged $^3\text{He}(n,p)t$ products (confined to the gas).